

Parametric Physics-Informed Neural Networks (PINN) for European Option Pricing: A Comparative Study on AAPL Market Data

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What is an option?

Stock



- piece of the company
- no expiry

Option



- right to buy a piece
- expires

How is it priced?

Stock



- S - stock price

Option



- S - stock price
- K - strike price
- V - Premium

Example with an option

- AAPL trading at \$270 (stock price)
- Buy an option for $K = \$270$ (strike price), $V = \$5$ (premium)

Scenario	Stock Outcome	Your Profit
Stock $\rightarrow$ \$320	Exercise	\$45
Stock $\rightarrow$ \$270	Don't Exercise	-\$5
Stock $\rightarrow$ \$0	Don't Exercise	-\$5

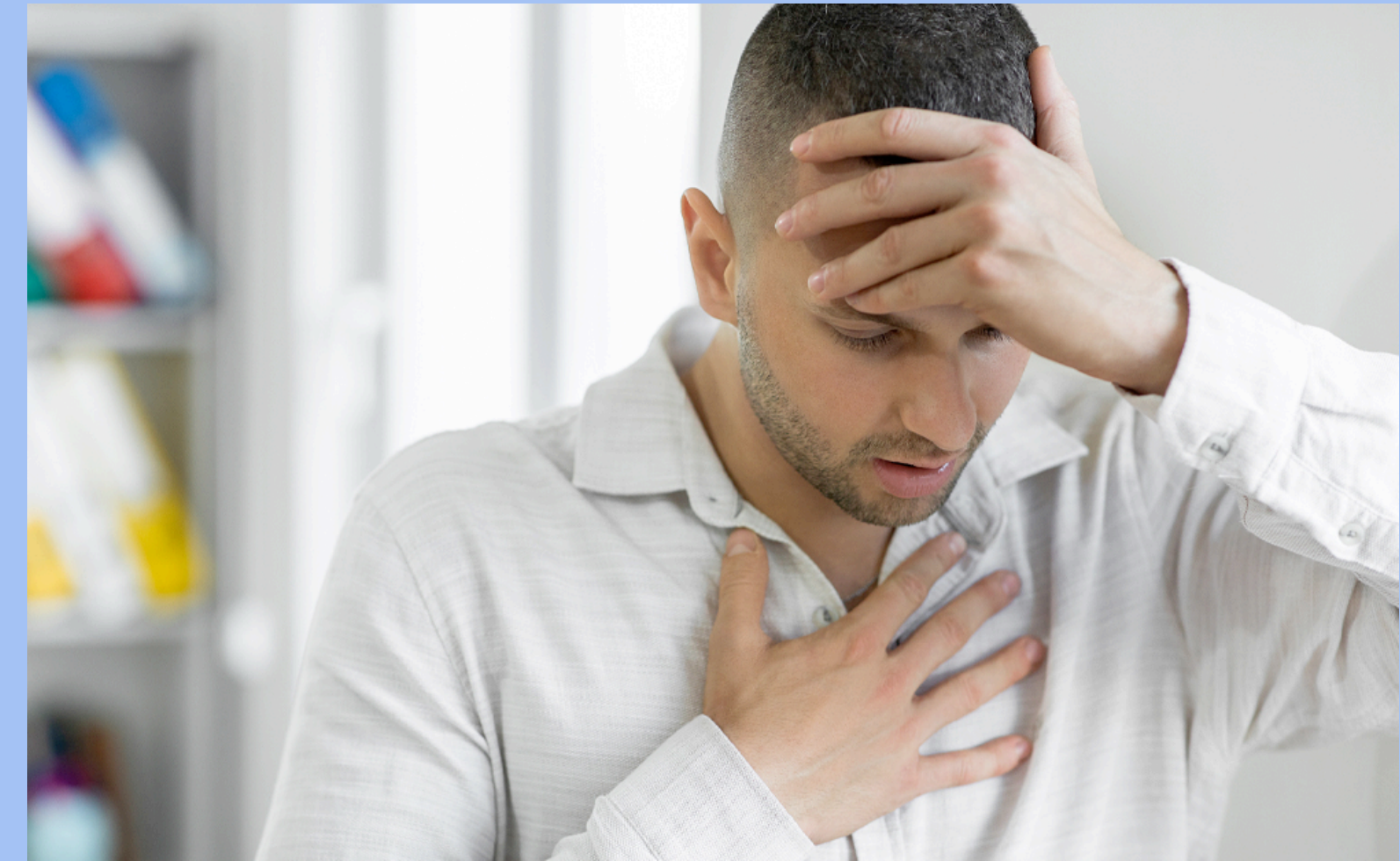
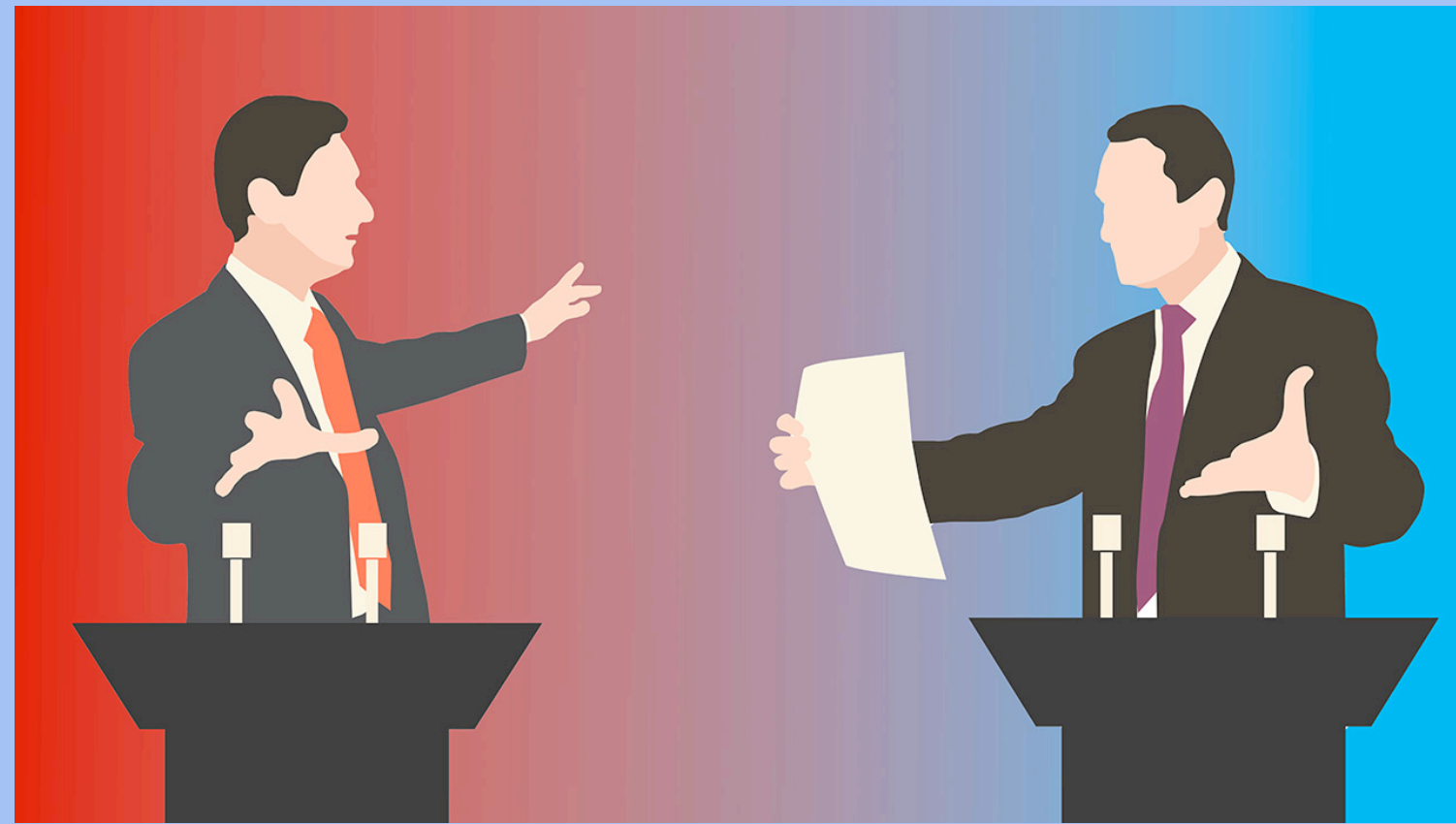
Upside is unlimited.
Downside is capped at V .

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Research Question:
What is the right V given S , K , t ?

Black-Scholes-Merton

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Black-Scholes-Merton

Assumptions:

- Constant volatility
- Normal distribution of changes
- Continuous trading
- No transaction costs

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$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Black-Scholes-Merton

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- S - stock price
- K - strike price
- t - time to expire
- σ - volatility (0.14)
- r - risk-free rate (0.04)
- V - option price (target)

Black-Scholes-Merton

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

...has a closed-form solution for European options

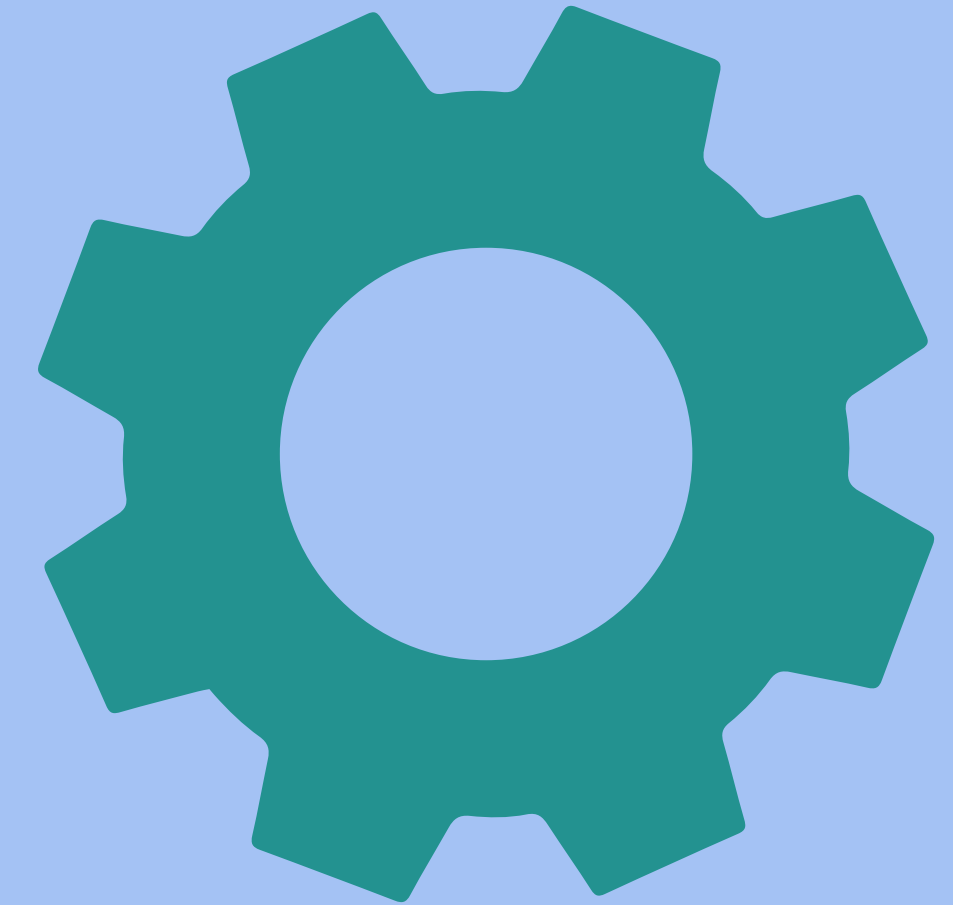
Why Neural Networks?



American options have
no closed solutions



BSM assumes
constant volatility



Grid-based numerical
methods do not scale
well

Why not just Deep Learning?

Regular Neural Networks

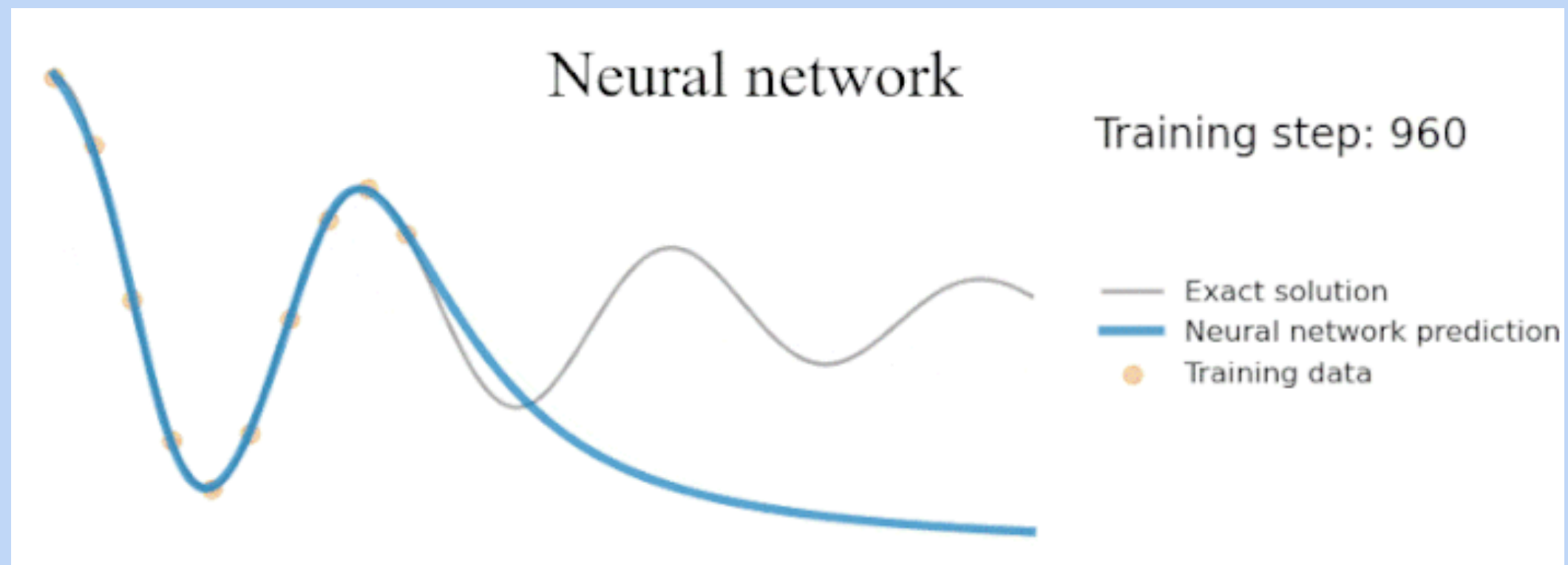
- Needs thousands of real market prices to train on
- Learns patterns in data but doesn't understand the math
- Extrapolates nonsense in data-sparse regions

Physics Informed Neural Networks

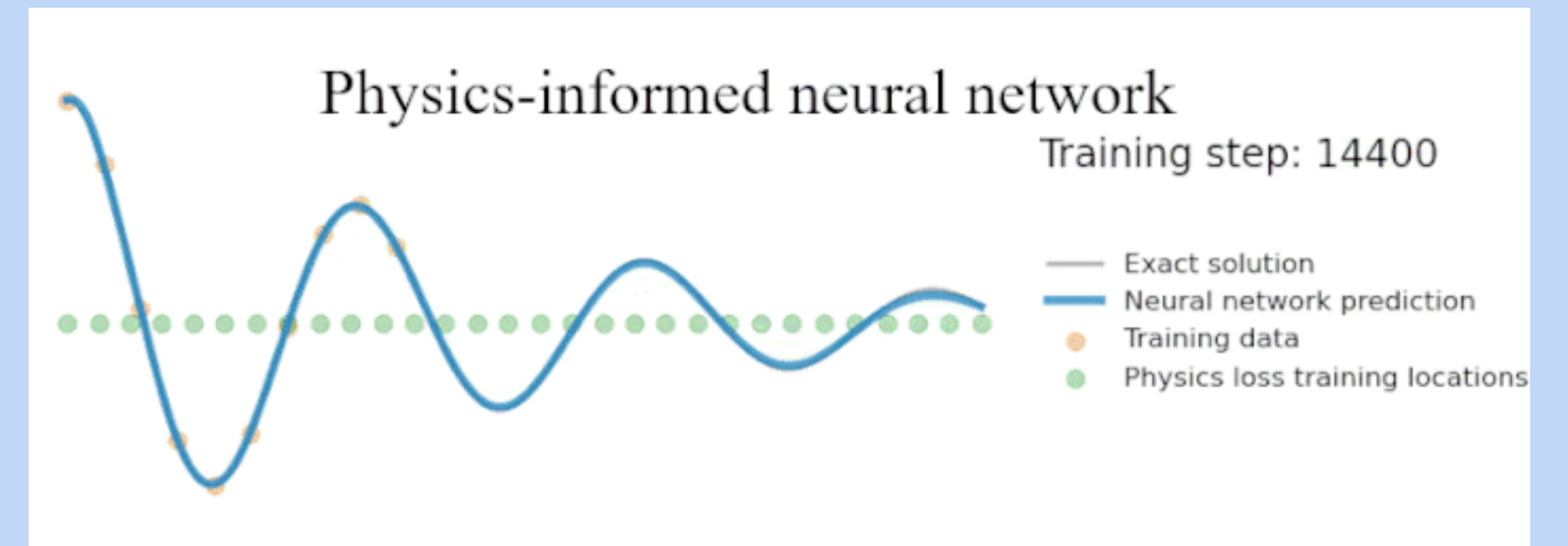
- Trained on synthetic points — needs zero real market prices
- Learns the BSM equation itself, not just patterns
- Generalizes correctly across the entire (S, t, K) domain

Why not just Deep Learning?

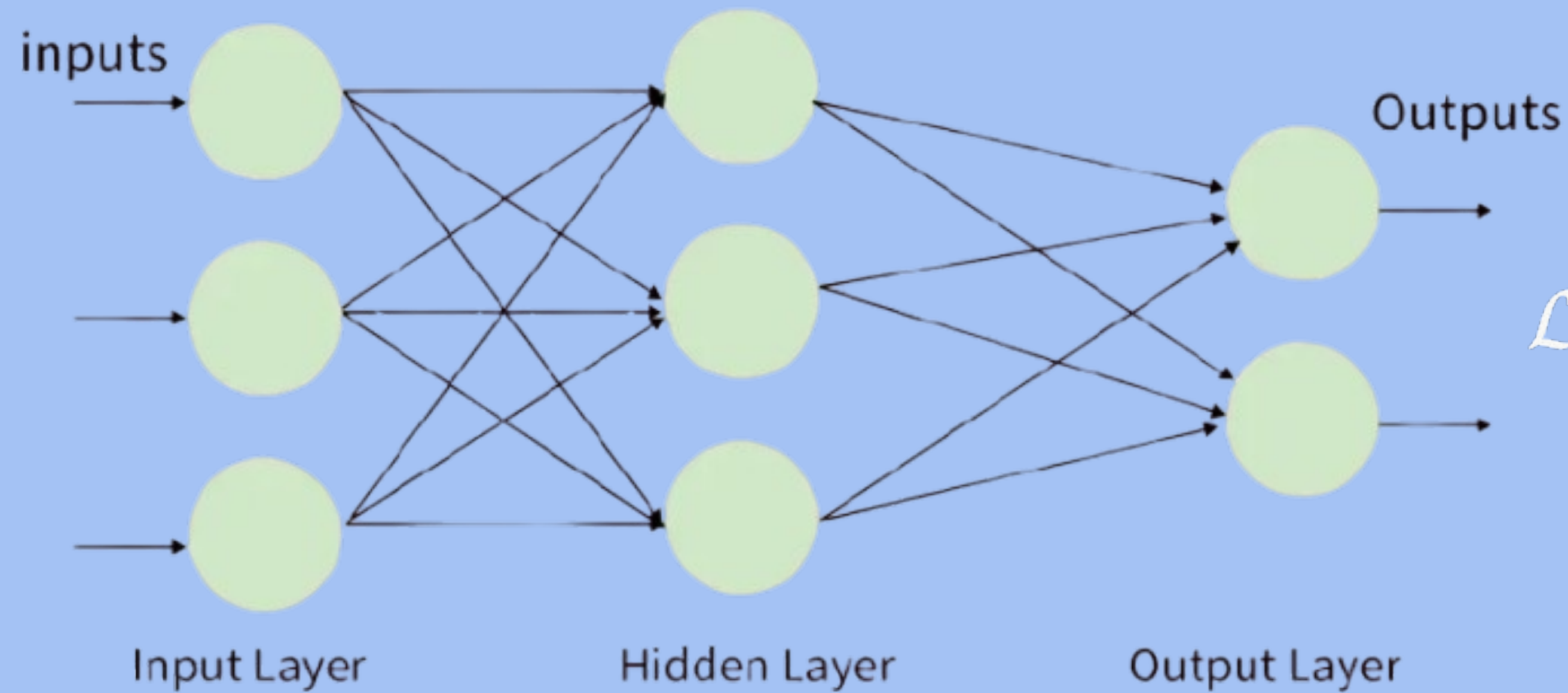
Regular Neural Networks



Physics Informed Neural Networks



What is a PINN?



$$\mathcal{L} = \underbrace{\text{PDE Loss}}_{\text{satisfy BSM equation}} + \underbrace{\text{Boundary Loss}}_{\text{correct payoff at expiry}}$$

PDE Loss - punishes every time Black-Scholes-Merton equation is violated

Boundary Loss - punishes every time option price at expiry doesn't match the known real payoff $\max(S-K, 0)$.

Our Contribution

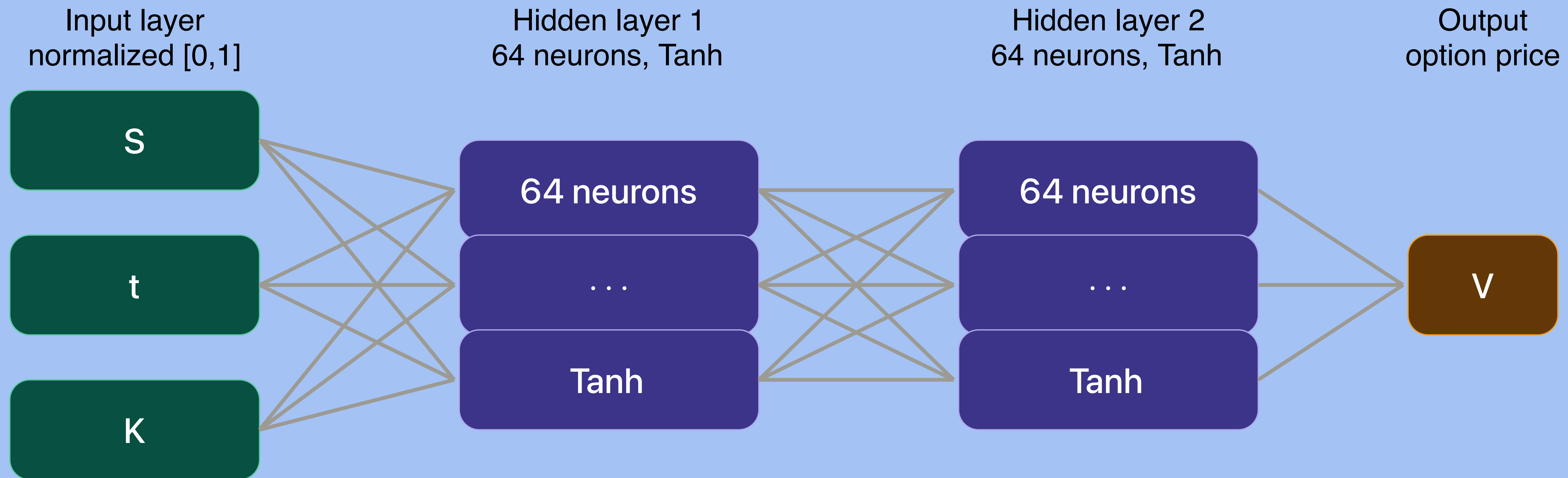
Previous PINN

- K is fixed
- One model per strike price (K)
- 10 different strikes = 10 different models
- Does NOT generalize well across the options chain

Parametric PINN

- K is an input to NN
- One model prices everything
- Feed any $(S, t, K) \rightarrow V$
- Continuous price surface

Architecture



Tanh

BSM needs second derivative, so ReLU is irrelevant

Float64

Second-order derivatives amplify floating point errors so double precision keeps training stable

Training setup

10,000

PDE collocation points

$S \in [170, 370]$ $t \in [0, 1]$ $K \in [220, 320]$

2,000

Boundary condition points

Sampled at $t = T$ (expiry)

10,000

Training epochs

Adam optimizer $lr = 0.001$

Calibrated Parameters

$r = 0.04$ — 13-week T-Bill yield

$\sigma = 0.14$ — 1yr AAPL historical vol

Hardware

NVIDIA RTX 6000 Ada (96GB)

float64 — double precision

Training Convergence



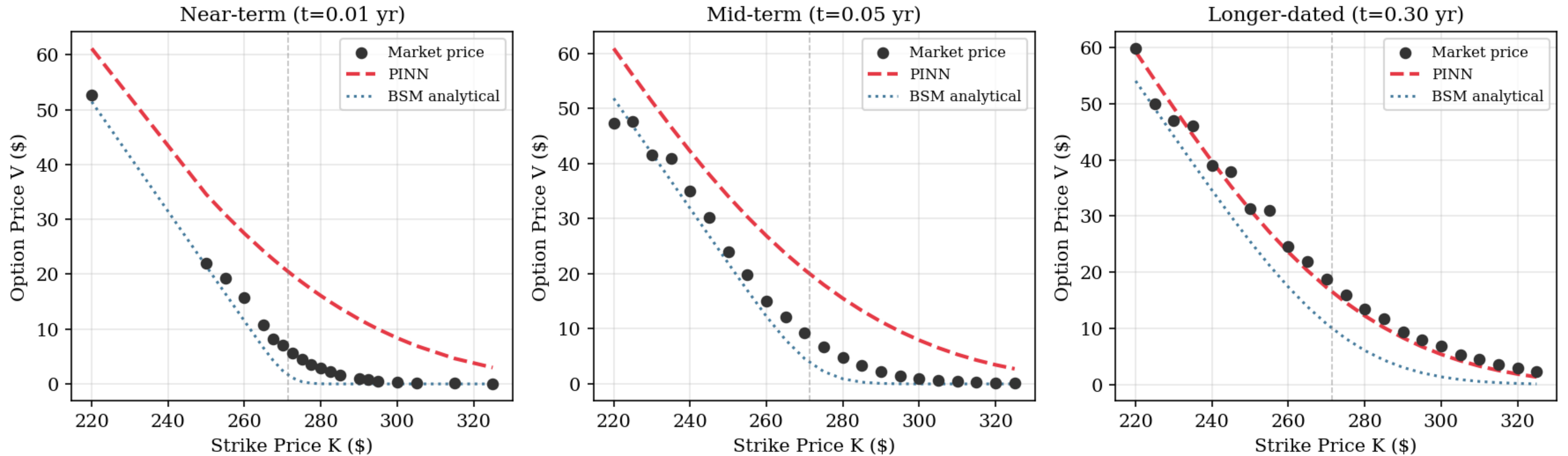
Results

Tenor	PINN Mae	BSM MAE	Winner
t=0.01 yr (~3 days)	\$10.96	\$2.17	BSM
t=0.05 yr (~3 weeks)	\$8.39	\$2.28	BSM
t=0.30 yr (~11 weeks)	\$1.52	\$5.64	PINN

- PINN beats 50-year-old exact formula for longer-dated options
- BBSM wins near expiry

Results

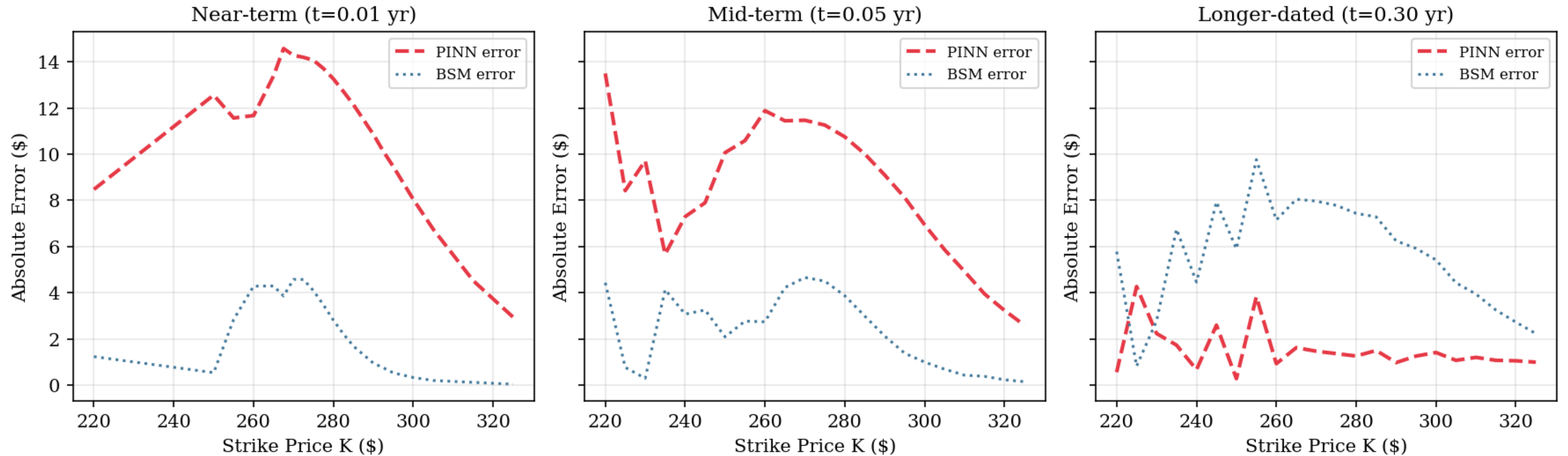
PINN vs. BSM Analytical vs. Market -- AAPL Call Options



PINN hugs the market dots at $t=0.30$ yr but diverges badly near expiry

Results

Absolute Pricing Error: PINN vs. BSM Analytical



BSM dominates PINN near expiry, but flips at $t=0.30$ yr where PINN error drops below BSM for the first time

Interpretation

✓ Far from expiry ($t = 0.30$ yr)

Price surface is smooth $\rightarrow$ network represents it well

$\rightarrow$ **PINN wins**

MAE \$1.52 vs BSM \$5.64

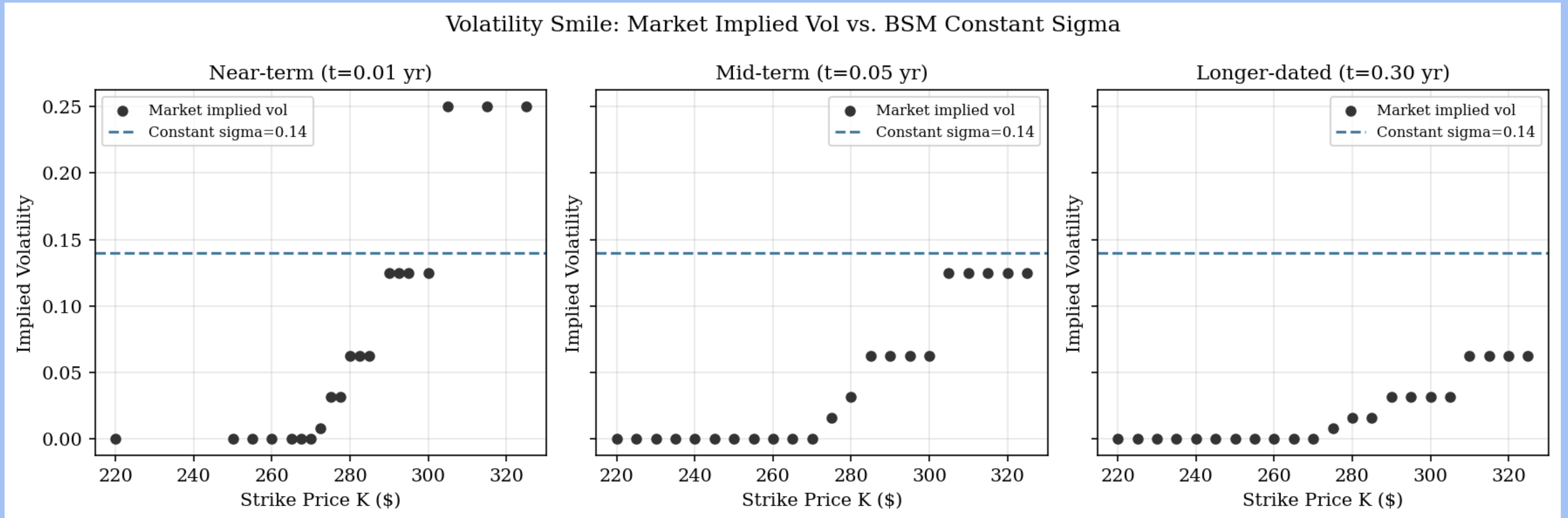
✗ Near expiry ($t = 0.01$ yr)

$\max(S-K, 0)$ becomes a step function — sharp cliff at $S=K$

Smooth Tanh networks blur the corner $\rightarrow$ large errors

This is the **sharp boundary problem**

Interpretation



We assume an equal volatility ($\sigma=0.14$), but real market has high volatility for OTM options

Social Implications & Future Work

Social Implications

- Institutional edge - faster pricing gives arbitrage advantage over slower participants
- Lower barrier - open source gives retail investors access to institutional-grade pricing
- Systemic risk - Correlated model failures during crashes.

Future Work

- Training longer with cosine LR schedule
- Inverse PINN (learn the volatility)
- Generalize to American options with non-fixed expiry date

THANK YOU!!

